Kingdom of Saudi Arabia
Ministry of Education
Umm Al-Qura University
031

المملكة العربية السعودية
وزارة التعليم
جامعة أم القرى
٠٣١

Project Report

Daily Calorie Needs Calculator

Team member :-

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Dr . Eman Alharbi

1. Introduction:

In today's world, monitoring calorie intake and nutrition has become an essential part of a healthy lifestyle. Many individuals seek to understand their daily energy needs based on their physical activity and body metrics. Honestly, the idea for this project stemmed from a personal need—we were planning to start following a healthy diet and found ourselves looking for a simple tool to calculate daily calorie requirements and macronutrient distribution. From there, the idea arose to build a tool that could help us—and others—organize food intake and track daily progress.

We designed a simple Bash script that calculates the individual's Total Daily Energy Expenditure (TDEE), Body Mass Index (BMI), and macronutrient distribution (proteins, carbohydrates, fats). In addition, the program allows users to track their daily calorie intake and compare it to their required daily amount. The goal of this project is to provide users with an easy-to-use tool that can be run via the terminal without the need for internet access or external applications.

2. Design and Implementation:

The project was implemented using the Bash scripting language due to its simplicity and ease of use. We utilized several commands learned during this course, while others were researched and understood to fit the program’s structure and improve its functionality.

3. Version Control and Collaboration using GitHub:

To ensure efficient collaboration and track our code development throughout the project, we used **GitHub** as our version control platform. Each team member worked on a separate branch, allowing us to develop features independently and later merge them into the main branch. This approach helped avoid code conflicts and preserved a clear history of all contributions.

3.1. Git Setup and Repository Connection:

We initialized a Git repository locally and connected it to GitHub using HTTPS. The connection allows each team member to push their changes remotely.

```
ubuntuu@ubuntuu-VirtualBox:~$ git clone https://github.com/Bas-87/Calories-Calculator.git
Cloning into 'Calories-Calculator'...
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (6/6), done.
Resolving deltas: 100% (1/1), done.
```

3.2. Branching Strategy:

Each team member created a personal branch named after their feature or name, such as Maha-featur, Batoul-Hallaq, etc.

This made the work modular and easy to manage.

```
mahanoorahmed@ubontu22:~/Calories-Calculator$ git checkout -b Maha-featur
Switched to a new branch 'Maha-featur'
```

3.3. Commits and Pushes:

As development progressed, each contributor committed their changes with descriptive messages that included their name and date for clear tracking. Afterwards, they pushed their updates to their respective branches.

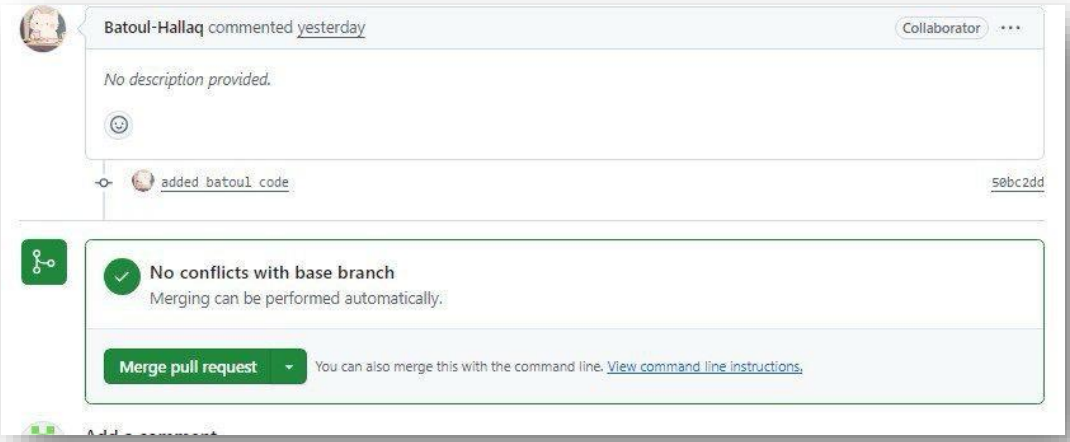
```
batoul@batoul-VirtualBox:~/Calories-Calculator$ git add .
batoul@batoul-VirtualBox:~/Calories-Calculator$ git commit -m "added batoul code"
[Batoul-Hallaq 50bc2dd] added batoul code
 1 file changed, 25 insertions(+)
 create mode 100644 batoul-code.sh
batoul@batoul-VirtualBox:~/Calories-Calculator$ git push origin Batoul-Hallaq
```

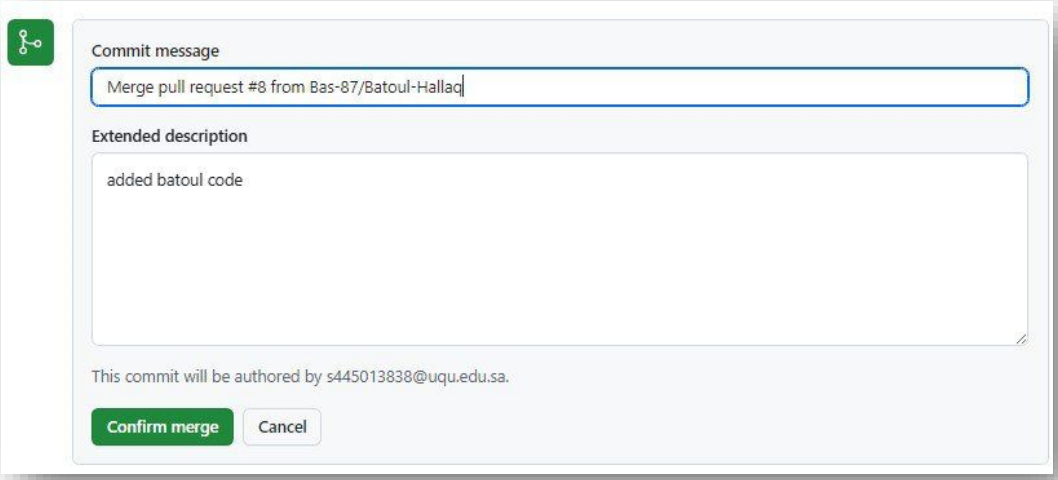
3.4. Merging and Finalization:

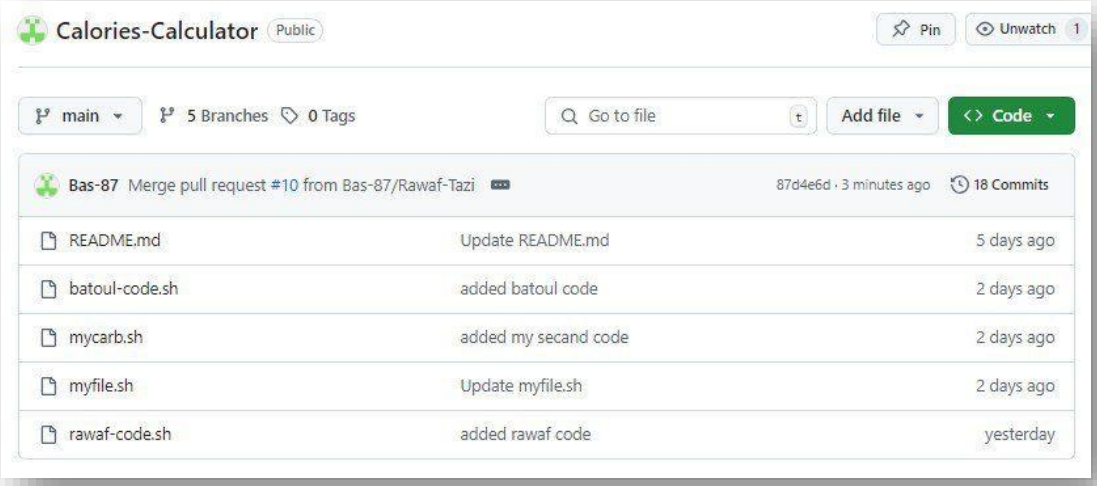
After reviewing the code in each branch, we planned to merge stable versions into the main branch. This ensured the final script included all working features.

Active branches

Branch	Updated	Check status	Behind	Ahead	Pull request
Rawaf-Tazi	35 minutes ago		0	1	#10
Maha-featur	7 hours ago		2	2	#9
Batoul-Hallaq	8 hours ago		0	1	#8
Basmah	13 hours ago		2	3	#1


Batoul-Hallaq commented yesterday
Collaborator
No description provided.
added batoul code
50bc2dd
No conflicts with base branch
Merging can be performed automatically.
Merge pull request
You can also merge this with the command line. View command line instructions.


Commit message
Merge pull request #8 from Bas-87/Batoul-Hallaq
Extended description
added batoul code
This commit will be authored by s445013838@uqu.edu.sa.
Confirm merge
Cancel


Calories-Calculator
Public
Pin
Unwatch 1
main
5 Branches
0 Tags
Go to file
t
Add file
Code
Bas-87 Merge pull request #10 from Bas-87/Rawaf-Tazi
87d4e6d · 3 minutes ago
18 Commits
README.md
Update README.md
5 days ago
batoul-code.sh
added batoul code
2 days ago
mycarb.sh
added my secand code
2 days ago
myfile.sh
Update myfile.sh
2 days ago
rawaf-code.sh
added rawaf code
yesterday

```
batoul@batoul-VirtualBox: ~/Calories-Calculator
batoul@batoul-VirtualBox:~$ git clone https://github.com/Bas-87/Calories-Calculator.git
Cloning into 'Calories-Calculator'...
remote: Enumerating objects: 18, done.
remote: Counting objects: 100% (18/18), done.
remote: Compressing objects: 100% (14/14), done.
remote: Total 18 (delta 4), reused 3 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (18/18), 7.01 KiB | 652.00 KiB/s, done.
Resolving deltas: 100% (4/4), done.
batoul@batoul-VirtualBox:~$ cd Calories-Calculator
batoul@batoul-VirtualBox:~/Calories-Calculator$ git checkout -b Batoul-Hallaq
Switched to a new branch 'Batoul-Hallaq'
batoul@batoul-VirtualBox:~/Calories-Calculator$ touch batoul-code.sh
batoul@batoul-VirtualBox:~/Calories-Calculator$ git add .
batoul@batoul-VirtualBox:~/Calories-Calculator$ git commit -m "added batoul code"
[Batoul-Hallaq 50bc2dd] added batoul code
 1 file changed, 25 insertions(+)
 create mode 100644 batoul-code.sh
batoul@batoul-VirtualBox:~/Calories-Calculator$ git push origin Batoul-Hallaq
Username for 'https://github.com': Batoul-Hallaq
Password for 'https://Batoul-Hallaq@github.com':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 636 bytes | 636.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'Batoul-Hallaq' on GitHub by visiting:
remote:      https://github.com/Bas-87/Calories-Calculator/pull/new/Batoul-Hallaq
remote:
To https://github.com/Bas-87/Calories-Calculator.git
 * [new branch]      Batoul-Hallaq -> Batoul-Hallaq
batoul@batoul-VirtualBox:~/Calories-Calculator$
```

```
ubuntuu@ubuntuu-VirtualBox:~$ git clone https://github.com/Bas-87/Calories-Calculator.git
Cloning into 'Calories-Calculator'...
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (6/6), done.
Resolving deltas: 100% (1/1), done.
ubuntuu@ubuntuu-VirtualBox:~$ cd Calories-Calculator
ubuntuu@ubuntuu-VirtualBox:~/Calories-Calculator$ git checkout -b Basmah
Switched to a new branch 'Basmah'
ubuntuu@ubuntuu-VirtualBox:~/Calories-Calculator$ touch myfile.sh
ubuntuu@ubuntuu-VirtualBox:~/Calories-Calculator$ git add .
ubuntuu@ubuntuu-VirtualBox:~/Calories-Calculator$ git commit -m " added my first code"
[Basmah 717c099]  added my first code
 1 file changed, 51 insertions(+)
 create mode 100755 myfile.sh
ubuntuu@ubuntuu-VirtualBox:~/Calories-Calculator$ git push origin Basmah
Username for 'https://github.com': Bas-87
Password for 'https://Bas-87@github.com':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 1021 bytes | 1021.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'Basmah' on GitHub by visiting:
remote:      https://github.com/Bas-87/Calories-Calculator/pull/new/Basmah
remote:
To https://github.com/Bas-87/Calories-Calculator.git
 * [new branch]      Basmah -> Basmah
ubuntuu@ubuntuu-VirtualBox:~/Calories-Calculator$
```

```
mahanoorahmed@ubontu22:~$ git clone https://github.com/Bas-87/Calories-Calculato
r.git
fatal: destination path 'Calories-Calculator' already exists and is not an empty
 directory.
mahanoorahmed@ubontu22:~$ cd Calories-Calculator
mahanoorahmed@ubontu22:~/Calories-Calculator$ git checkout -b Maha-featur
Switched to a new branch 'Maha-featur'
mahanoorahmed@ubontu22:~/Calories-Calculator$ touch mycarb.sh
mahanoorahmed@ubontu22:~/Calories-Calculator$ git add .
mahanoorahmed@ubontu22:~/Calories-Calculator$ git commit -m "added maha code"
On branch Maha-featur
nothing to commit, working tree clean
mahanoorahmed@ubontu22:~/Calories-Calculator$ git push origin Maha-featur
Username for 'https://github.com': MahaNr24
Password for 'https://MahaNr24@github.com':
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 2 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (13/13), 4.52 KiB | 4.52 MiB/s, done.
Total 13 (delta 2), reused 6 (delta 1), pack-reused 0
remote: Resolving deltas: 100% (2/2), done.
remote:
remote: Create a pull request for 'Maha-featur' on GitHub by visiting:
remote:      https://github.com/Bas-87/Calories-Calculator/pull/new/Maha-featur
remote:
To https://github.com/Bas-87/Calories-Calculator.git
 * [new branch]      Maha-featur -> Maha-featur
mahanoorahmed@ubontu22:~/Calories-Calculator$
```

```
rawaf@ubuntu22: ~/Desktop/Calories-Calculator
rawaf@ubuntu22:~/Desktop$ git clone https://github.com/Bas-87/Calories-Calculator.git
Cloning into 'Calories-Calculator'...
remote: Enumerating objects: 18, done.
remote: Counting objects: 100% (18/18), done.
remote: Compressing objects: 100% (14/14), done.
remote: Total 18 (delta 4), reused 3 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (18/18), 7.01 KiB | 1.40 MiB/s, done.
Resolving deltas: 100% (4/4), done.
rawaf@ubuntu22:~/Desktop$ cd Calories-Calculator
rawaf@ubuntu22:~/Desktop/Calories-Calculator$ git checkout -b Rawaf-Tazi
Switched to a new branch 'Rawaf-Tazi'
rawaf@ubuntu22:~/Desktop/Calories-Calculator$ touch rawaf-code.sh
rawaf@ubuntu22:~/Desktop/Calories-Calculator$ git add .
rawaf@ubuntu22:~/Desktop/Calories-Calculator$ git commit -m "added rawaf code"
[Rawaf-Tazi 83a3304] added rawaf code
 Committer: Rawaf <rawaf@ubuntu22.myguest.virtualbox.org>
Your name and e-mail address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 1 file changed, 53 insertions(+)
 create mode 100644 rawaf-code.sh
rawaf@ubuntu22:~/Desktop/Calories-Calculator$ git push origin Rawaf-Tazi
Username for 'https://github.com': Rawaf-Tazi
Password for 'https://Rawaf-Tazi@github.com':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 946 bytes | 946.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
remote:
remote: Create a pull request for 'Rawaf-Tazi' on GitHub by visiting:
remote:      https://github.com/Bas-87/Calories-Calculator/pull/new/Rawaf-Tazi
remote:
To https://github.com/Bas-87/Calories-Calculator.git
 * [new branch]      Rawaf-Tazi -> Rawaf-Tazi
rawaf@ubuntu22:~/Desktop/Calories-Calculator$
```

Github link: https://github.com/Bas-87/Calories-Calculator

4. The program performs several interconnected tasks:

4.1. Collecting user data: name, gender, age, weight, height, and activity level.

```bash
#!/bin/bash
# === Basmah's Function: Collect Input + Calculate BMR & TDEE ===
calculate_bmr_tdee() {
    echo "==========================================="
    echo "🌟 Welcome to the Daily Calorie Calculator 🌟"
    echo "==========================================="
    echo ""

    # Get user's data
    read -p "Enter your name: " name
    read -p "Enter your gender (male/female): " gender
    read -p "Enter your age: " age
    read -p "Enter your weight in kg: " weight
    read -p "Enter your height in cm: " height

    # Prompt for activity level
    echo "activity level:"
    echo "1. Sedentary (little or no exercise)"
    echo "2. Lightly active (1-3 days/week)"
    echo "3. Moderately active (3-5 days/week)"
    echo "4. Very active (6-7 days/week)"
    echo "5. Super active (intense exercise/twice daily)"
    read -p "Enter activity level [1-5]: " activity

    # Determine activity factor based on user's input
    case $activity in
        1) factor=1.2 ;;
        2) factor=1.375 ;;
        3) factor=1.55 ;;
        4) factor=1.725 ;;
        5) factor=1.9 ;;
        *) echo "Invalid input. Defaulting to sedentary."; factor=1.2 ;;
```

```
===========================================
🌟 Welcome to the Daily Calorie Calculator 🌟
===========================================

Enter your name: Rori
Enter your gender (male/female): female
Enter your age: 20
Enter your weight in kg: 49
Enter your height in cm: 150
activity level:
1. Sedentary (little or no exercise)
2. Lightly active (1-3 days/week)
3. Moderately active (3-5 days/week)
4. Very active (6-7 days/week)
5. Super active (intense exercise/twice daily)
Enter activity level [1-5]: 2
```

4.2. Calculating BMR and TDEE: using the Mifflin-St Jeor equations, which are considered among the most accurate for estimating basal metabolic rate.

```bash
    # Calculate BMR based on gender and user data
    if [ "$gender" -eq "male" ] 2>/dev/null
    then
        BMR=$(echo "$weight $height $age" | awk '{print (10 * $1) + (6.25 * $2) - (5 * $3) + 5}')
    else
        BMR=$(echo "$weight $height $age" | awk '{print (10 * $1) + (6.25 * $2) - (5 * $3) - 161}')
    fi

    # Calculate TDEE by multiplying BMR with activity factor
    TDEE=$(echo "$BMR $factor" | awk '{print $1 * $2}')

    # Output the results
    echo ""
    echo "Hello $name! Based on your data:"
    echo "Your Total Daily Energy Expenditure (TDEE) is: $TDEE kcal/day"
}

# Call the function to run
calculate_bmr_tdee
```

```
Hello Rori! Based on your data:
Your Total Daily Energy Expenditure (TDEE) is: 1603.94 kcal/day
```

4.3. Calculating macronutrient distribution: TDEE is divided into 20% protein, 50% carbohydrates, and 30% fats.

Open

mycarb.sh
~/Calories-Calculator

```bash
#!/bin/bash
# === Maha's Function: Calculate Macronutrients ===
# Protein: 20% of TDEE ÷ 4 = g
# Carbs: 50% ÷ 4 = g
# Fat: 30% ÷ 9 = g
echo ""
echo "======================================"
echo "  🍔 Calculate Macronutrients 🥗"
echo "======================================"
echo ""
PROTEIN=$(awk -v t="$TDEE" 'BEGIN { printf "%.1f", (t * 0.20) / 4 }')
CARBS=$(awk -v t="$TDEE" 'BEGIN { printf "%.1f", (t * 0.50) / 4 }')
FATS=$(awk -v t="$TDEE" 'BEGIN { printf "%.1f", (t * 0.30) / 9 }')

# Output the results
echo "BMR: $BMR kcal/day"
echo "Protein: $PROTEIN g/day"
echo "Carbs: $CARBS g/day"
echo "Fats: $FATS g/day"
```

```
======================================
  🍔 Calculate Macronutrients 🥗
======================================

BMR: 1166.5 kcal/day
Protein: 80.2 g/day
Carbs: 200.5 g/day
Fats: 53.5 g/day
```

4.4. Calculating BMI: to provide an assessment of the user's current weight status.

```bash
#!/bin/bash
#=== Batoul's Function: Body Mass Index (BMI) Calculator ===
echo ""
echo "======================================"
echo "⚖️  Body Mass Index (BMI) Calculator ⚖️"
echo "======================================"
echo ""
# Convert height from centimeters to meters
height_m=$(echo "scale=2; $height / 100" | bc)
# Calculate BMI: BMI = weight (kg) / height (m)^2
bmi=$(echo "scale=2; $weight / ($height_m * $height_m)" | bc)
echo "$name, based on height and weight, your BMI is: $bmi"
# Determine BMI category using awk
if awk "BEGIN {exit !($bmi < 18.5)}"; then
    echo "Underweight"
elif awk "BEGIN {exit !($bmi >= 18.5 && $bmi < 25)}"; then
    echo "Normal weight"
elif awk "BEGIN {exit !($bmi >= 25 && $bmi < 30)}"; then
    echo "Overweight"
else
    echo "Obese"
fi
```

```
======================================
⚖️ Body Mass Index (BMI) Calculator ⚖️
======================================

Rori, based on height and weight, your BMI is: 21.77
Normal weight
```

4.5. Tracking daily calorie intake: the program allows users to interactively input meals and determines whether they have met or exceeded their daily calorie goal.

rawaf-code.sh
~/Desktop/Calories-Calculator

```
#!/bin/bash
#=== Rawaf's Function:Track daily calories===
echo ""
echo "========================================"
echo "      ◆ Track daily calories ◆ "
echo "========================================"
echo ""
total_intake=0

while true; do
    read -p "Do you want to add a meal/snack? (yes/no): " choice
    if [ "$choice" = "no" ] 2>/dev/null
    then
        break
    fi
    read -p "Enter the number of calories for this meal: " calories
    total_intake=$(echo "$total_intake + $calories" | bc)
    echo "Current total intake: $total_intake kcal"
done

#Compared to required calories
echo ""
echo "=== Intake Summary ==="
echo "Total consumed today: $total_intake kcal"
echo "Daily need: $TDEE kcal"

```

```
========================================
      ◆ Track daily calories ◆
========================================

Do you want to add a meal/snack? (yes/no): yes
Enter the number of calories for this meal: 685
Current total intake: 685 kcal
Do you want to add a meal/snack? (yes/no): yes
Enter the number of calories for this meal: 845
Current total intake: 1530 kcal
Do you want to add a meal/snack? (yes/no): yes
Enter the number of calories for this meal: 174
Current total intake: 1704 kcal
Do you want to add a meal/snack? (yes/no): no

=== Intake Summary ===
Total consumed today: 1704 kcal
Daily need: 1603.94 kcal
You exceeded your daily calorie limit!
```

4.6. Saving results to a file: all data is saved to a timestamped text file for tracking purposes.

rawaf-code.sh
~/Desktop/Calories-Calculator

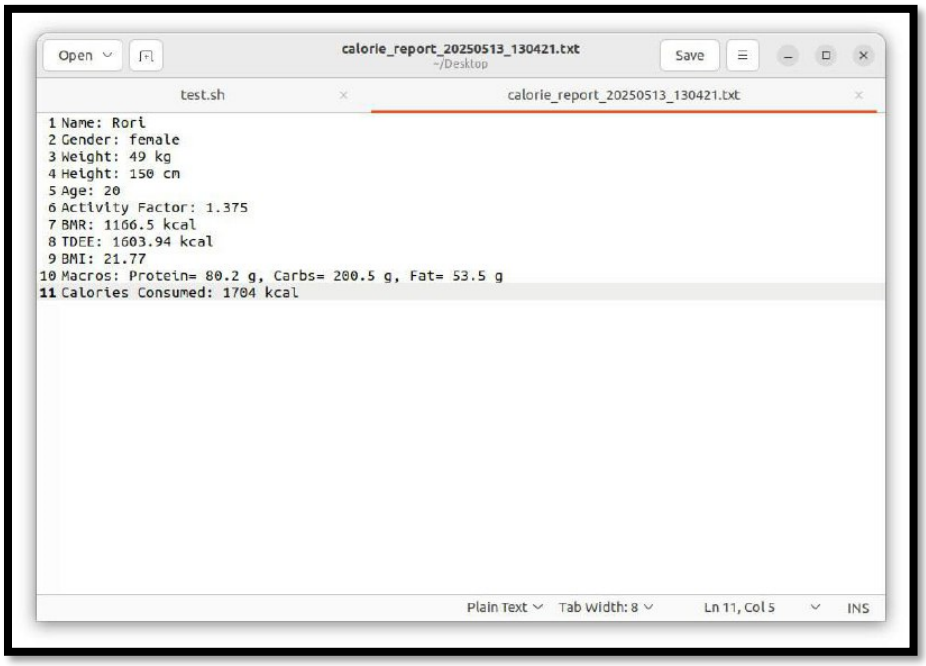
```
if [ $(echo $total_intake > $TDEE | bc -l) ]; then
    echo "You exceeded your daily calorie limit!"
elif [ $(echo $total_intake < $TDEE | bc -l) ]; then
    echo "You're still under your daily calorie needs."
else
    echo "You met your exact daily requirement!"
fi

#Save results
filename="calorie_report_$(date +%Y%m%d_%H%M%S).txt"
{
    echo "Name: $name"
    echo "Gender: $gender"
    echo "Weight: $weight kg"
    echo "Height: $height cm"
    echo "Age: $age"
    echo "Activity Factor: $factor"
    echo "BMR: $BMR kcal"
    echo "TDEE: $TDEE kcal"
    echo "BMI: $bmi"
    echo "Macros: Protein= $PROTEIN g, Carbs= $CARBS g, Fat= $FATS g"
    echo "Calories Consumed: $total_intake kcal"
} > "$filename"

echo ""
echo "Results saved to $filename"
```

```
Results saved to calorie_report_20250513_130421.txt
rawaf@ubuntu22:~/Desktop$
```

The saved file:-



4.7. Starter file: Call all codes in order.

```
# starting Basma's code: Collect Input + Calculate BMR & TDEE
source myfile.sh
# Maha's code: Calculate Macronutrients
source mycarb.sh
# Batoul's code: Body Mass Index (BMI) Calculator
source batoul-code.sh
# Rawaf's code: Track daily calories
source rawaf-code.sh
```

- Because our codes share variables and are closely linked to each other, we use "source" which means running the codes in one session and keeping the variables and outputs.

```
batoul-VirtualBox:~/Calories-Calculator$ bash main.sh
```

- When running the main file, we have to use the word "bash" because it supports "source".

The entire program:

We chose this modular and interactive approach to ensure ease of use, better maintainability, and efficient task distribution among team members, while also offering a flexible user experience.

```
============================================
🌟 Welcome to the Daily Calorie Calculator 🌟
============================================

Enter your name: Rori
Enter your gender (male/female): female
Enter your age: 20
Enter your weight in kg: 49
Enter your height in cm: 150
activity level:
1. Sedentary (little or no exercise)
2. Lightly active (1-3 days/week)
3. Moderately active (3-5 days/week)
4. Very active (6-7 days/week)
5. Super active (intense exercise/twice daily)
Enter activity level [1-5]: 2

Hello Rori! Based on your data:
Your Total Daily Energy Expenditure (TDEE) is: 1603.94 kcal/day

====================================
  🍔 Calculate Macronutrients 🥗
====================================

BMR: 1166.5 kcal/day
Protein: 80.2 g/day
Carbs: 200.5 g/day
Fats: 53.5 g/day

=======================================
⚖️ Body Mass Index (BMI) Calculator ⚖️
=======================================

Rori, based on height and weight, your BMI is: 21.77
Normal weight

=======================================
      🔹 Track daily calories 🔹
=======================================

Do you want to add a meal/snack? (yes/no): yes
Enter the number of calories for this meal: 685
Current total intake: 685 kcal
Do you want to add a meal/snack? (yes/no): yes
Enter the number of calories for this meal: 845
Current total intake: 1530 kcal
Do you want to add a meal/snack? (yes/no): yes
Enter the number of calories for this meal: 174
Current total intake: 1704 kcal
Do you want to add a meal/snack? (yes/no): no

=== Intake Summary ===
Total consumed today: 1704 kcal
Daily need: 1603.94 kcal
You exceeded your daily calorie limit!

Results saved to calorie_report_20250516_062449.txt
batoul@batoul-VirtualBox:~$
```

5. Testing and Results:

The program was tested using sample data for a user named "Rori", a 20-year-old female weighing 49 kg, with a height of 150 cm and a light activity level. The program returned the following values:

BMR: 1166.5 kcal/day

TDEE: 1603.94 kcal/day

BMI: 21.77 (Normal weight)

Protein: 80.2 g/day

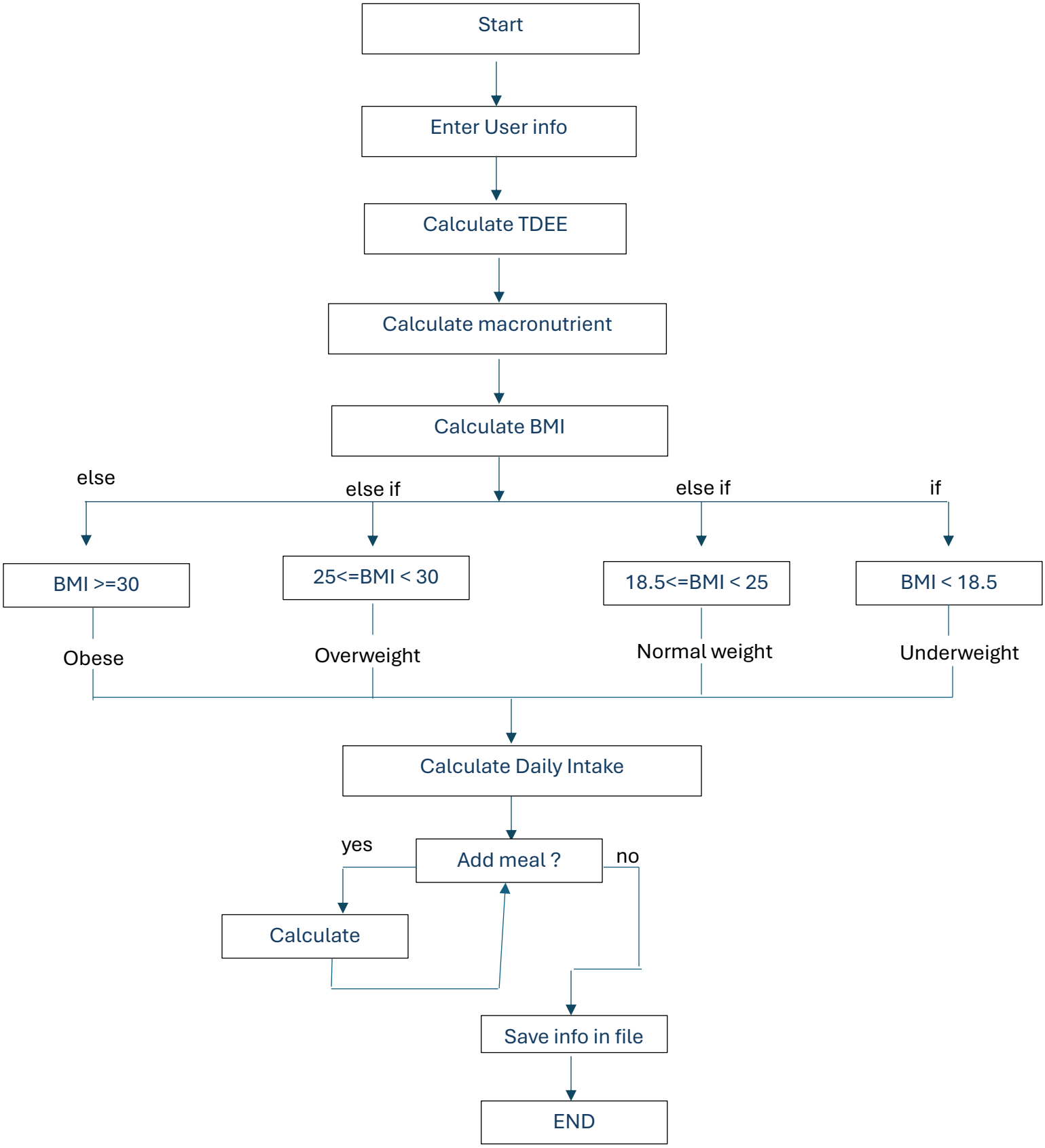
Carbohydrates: 200.5 g/day

Fats: 53.5 g/day

The calorie tracking feature was also tested by inputting three meals totaling 1704 kcal. The program correctly identified that the user had exceeded their daily calorie limit and provided an appropriate warning, while also saving the results in a file.

The script worked correctly from start to finish with no apparent logical or programming errors. However, there is room for further development and improvement.

6. Flowchart :


Start
Enter User info
Calculate TDEE
Calculate macronutrient
Calculate BMI
else
else if
else if
if
BMI >=30
25<=BMI < 30
18.5<=BMI < 25
BMI < 18.5
Obese
Overweight
Normal weight
Underweight
Calculate Daily Intake
yes
Add meal ?
no
Calculate
Save info in file
END

7. Conclusion:

This project successfully achieved its goal of providing a simple and practical tool for calculating calorie needs and monitoring daily intake. It integrated multiple concepts such as health-related formulas, interactive computation, and structured result storage.

8. Challenges faced:

- Limitations of Bash in handling precise decimal calculations and floating-point operations.

- Lack of a graphical user interface, which may make the tool less accessible for some users.

- Therefore, a possible future enhancement would be to reimplement the program using a more advanced programming language like Python or Java, which support GUI design and more robust numeric processing.

9. References:

- https://tdeecalculator.net/?utm_source=chatgpt.com
- https://www.elconsolto.com/calculator/bmr
- https://youtu.be/uakI44OUQw?si=-PsdraDGCgnti7or